第八章 主存管理

1. **Explain the difference between internal and external fragmentation.**

External: free gaps between allocated chunks

Internal: don't need all memory within allocated chunks

2. **Given five memory partitions of 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB (in order), how would each of the first-fit, best-fit, and worst-fit algorithms place processes of 212 KB, 417 KB, 112 KB, and 426 KB (in order)? Which algorithm makes the most efficient use of memory?**

First-fit: ① 212KB → 500, Left: 288

② 417KB → 600, Left: 183

③ 112KB → 288, Left: 176

④ 426KB → can't allocate

Best-fit: ① 212KB → 300, Left: 88

② 417KB → 500, Left: 83

③ 112KB → 200, Left: 88

④ 426KB → 600, Left: 174

Worst-fit: ① 212KB → 600, Left: 388

② 417KB → 500, Left: 83

③ 112KB → 388, Left: 276

④ 426KB → can't allocate

So, best-fit is most efficient.

3. Consider a paging system with the page table stored in memory.

a. If a memory reference takes 200 nanoseconds, how long does a paged memory reference take?

b. If we add TLBs, and 75 percent of all page-table reference are found in the TLBs, what is the effective memory reference time?(Assume that finding a page-table entry in the TLBs takes zero time, if the entry is there)

a. ① Read the page-table entry from memory: 200ns

② Read the word from memory: 200ns

∴ Need 400ns.

b. ① Hit: 200ns

② Miss: 400ns

∴ Total: $0.75 \times 200 + 0.25 \times 400 = 250ns$